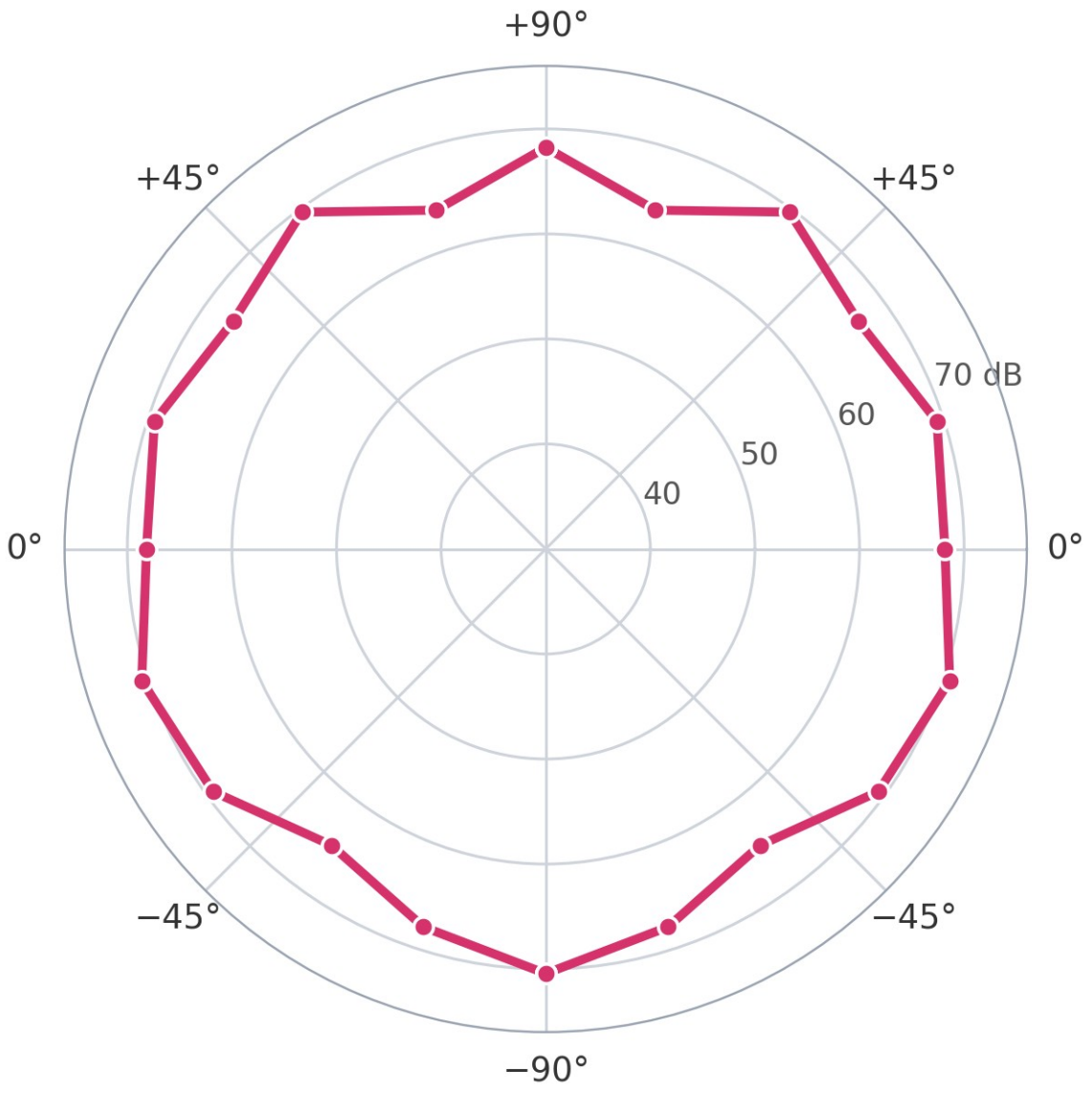


Chasing a round polar

Our chamber in its current set-up (5 Oct 2026): eleven microphones on a ring around the source, absorbing material on floor, ceiling and walls.



THE PROBLEM

Our polar plot is jagged

- We measured a propeller with the microphone arc in a plane parallel to the propeller disc.
- A propeller is symmetric around its axis, so every microphone should read about the same.
- **We expected a smooth, nearly round polar. We observed a jagged outline.**

6.7 dB

between the loudest and quietest of the eleven microphones

31 Aug 2026, 20–560 Hz, motor at 2000 µs. Radial axis 30–76 dB, the same as in the report.

THE FIX

What we did: every microphone, one loudspeaker



Our loudspeaker

tones, 62 Hz–16 kHz



UMIK-2 microphone, capsule faces the loudspeaker

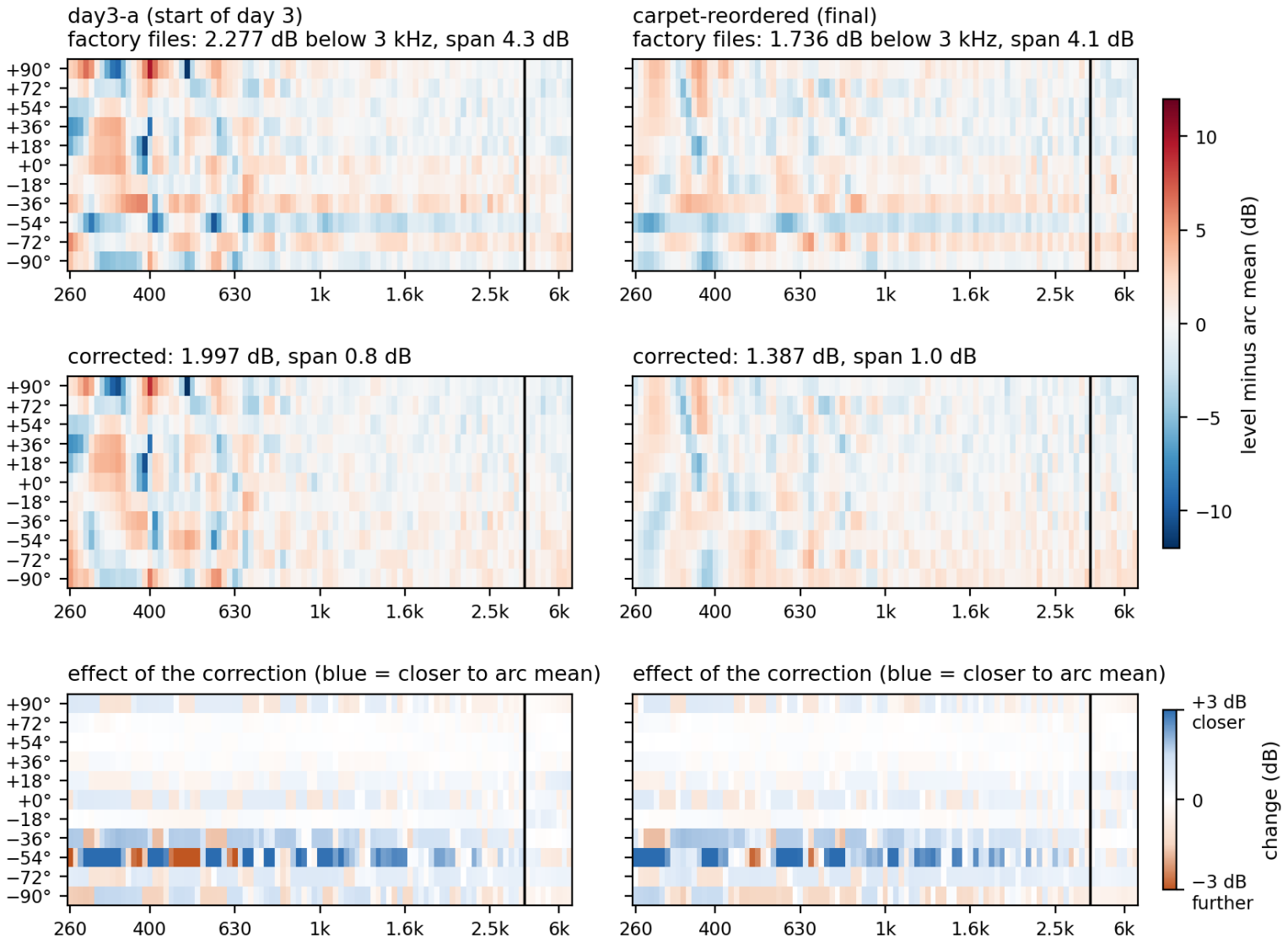
- 97 tones per microphone, 62 Hz to 16 kHz, 12 per octave (a stepped sweep)
- One tone at a time: 0.7 s to settle, then 2 s recorded (6 s below 250 Hz)
- Microphones swapped in turn: same position, same loudspeaker, same level
- One microphone measured three times as the reference

4.01 → 0.02 dB

spread across the eleven microphones for the same sound

7 of 11 factory files were off by 0.8–2.6 dB on average

Calibration session 16 Sep 2026, loudspeaker amplitude 0.03 of full scale. UMIK-2 photo: miniDSP product brief.



Frequency in Hz (k = kHz; the black line marks the omitted 3–5 kHz). Top of each map = +90°.

Two captures from 25 Sep 2026: the start of day 3 (left) and the final configuration (right).

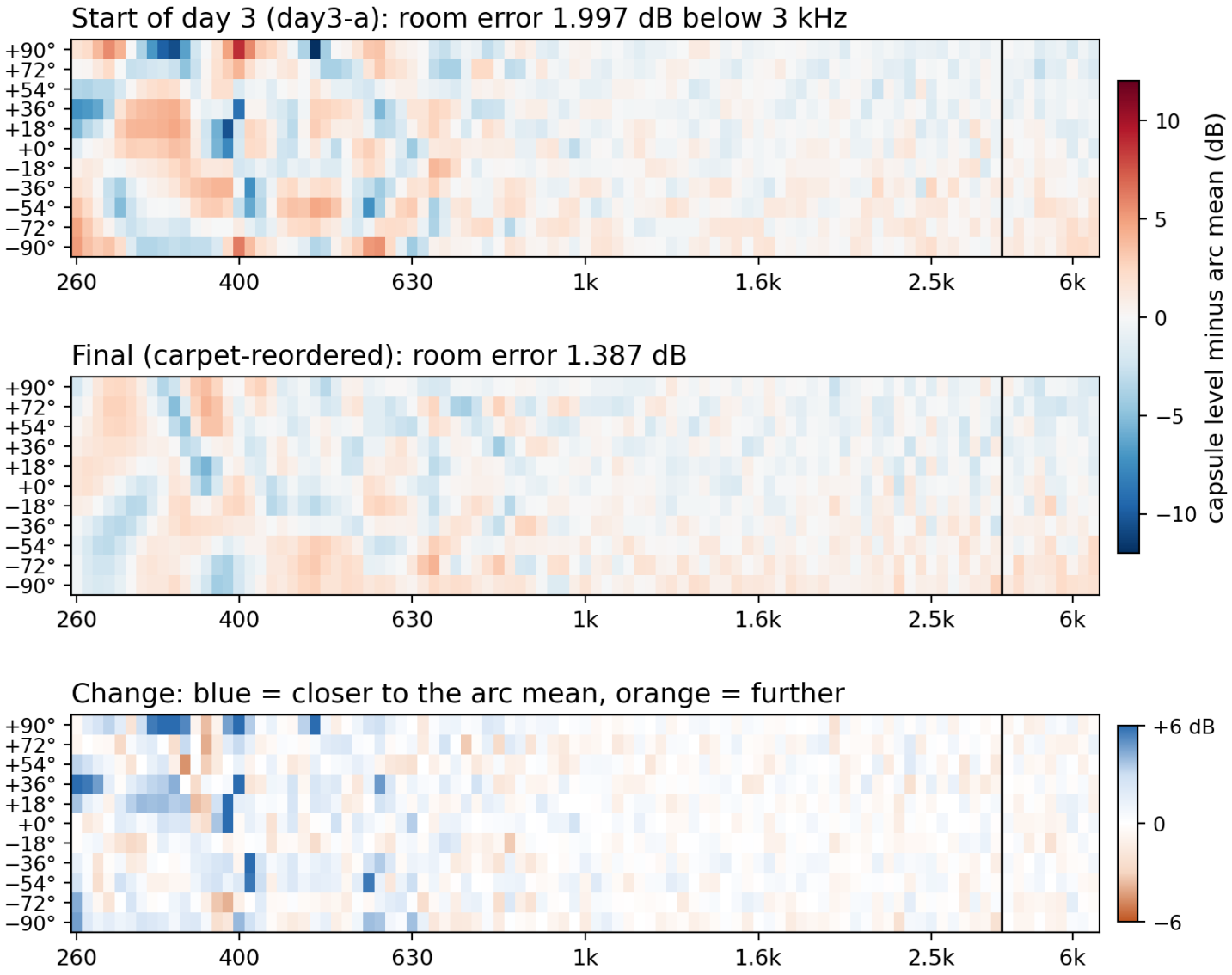
THE RESULT

The correction removes the microphone offsets

- **Top**: factory files. **Middle**: corrected. **Bottom**: what the correction changed.
- A stripe across all frequencies is one microphone reading too loud or too quiet. The middle row has none.
- The room pattern stays: whole rows move, not the frequency structure.

4.3 → 0.8 dB

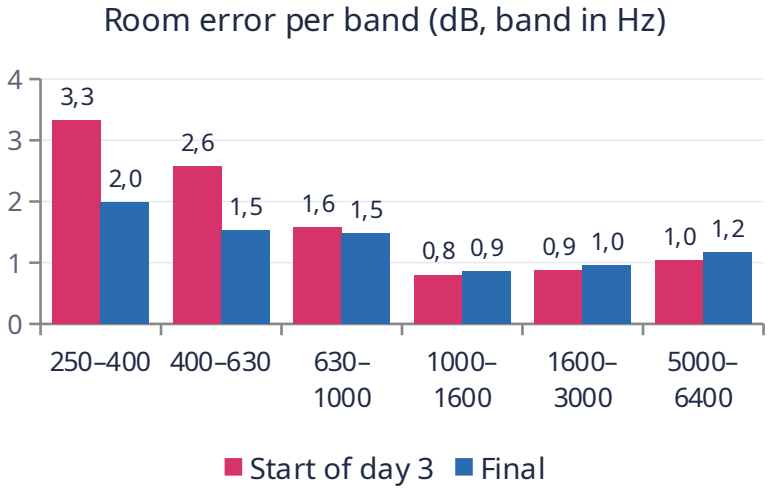
gap between the loudest and quietest microphone for the same sound: factory files → corrected (start of day 3; final state 4.1 → 1.0 dB)



Frequency in Hz (k = kHz; black line: omitted 3–5 kHz). Top of each map = +90°.

THE CHAMBER

The treatments flatten the low end



2.00 → 1.39 dB

room error below 3 kHz

We improved the low end (below 630 Hz) by 1.2 dB: the error fell from 3.0 to 1.8 dB, 40 % lower. Above 1 kHz only marginal changes.

25 Sep 2026. The loudspeaker was moved during the day and returned to about its starting point; the position was not documented. Room error: rms difference of the eleven microphones from their average.

Band	Limit	Tones	Start of day 3		Final	
Hz	± dB	per band	worst mic, dB	cells in limit	worst mic, dB	cells in limit
250	1.5	3	**6.81**	36 %	**1.88**	52 %
315	1.5	8	**3.46**	28 %	**0.89**	56 %
400	1.5	8	**3.46**	41 %	**1.41**	59 %
500	1.5	8	**1.93**	41 %	**1.99**	59 %
630	1.5	8	**0.93**	43 %	**1.49**	61 %
800	1.0	8	**0.87**	60 %	**1.12**	55 %
1000	1.0	9	**1.08**	71 %	**0.57**	73 %
1250	1.0	8	**0.54**	83 %	**1.02**	73 %
1600	1.0	8	**0.69**	75 %	**1.15**	76 %
2000	1.0	8	**0.66**	84 %	**0.77**	82 %
2500	1.0	8	**0.82**	64 %	**0.75**	65 %
5000*	1.0	4	**1.24**	73 %	**1.23**	64 %
6300*	1.5	5	**1.59**	78 %	**1.77**	76 %

THE EVALUATION

Band by band against the ISO tolerance

Colour key



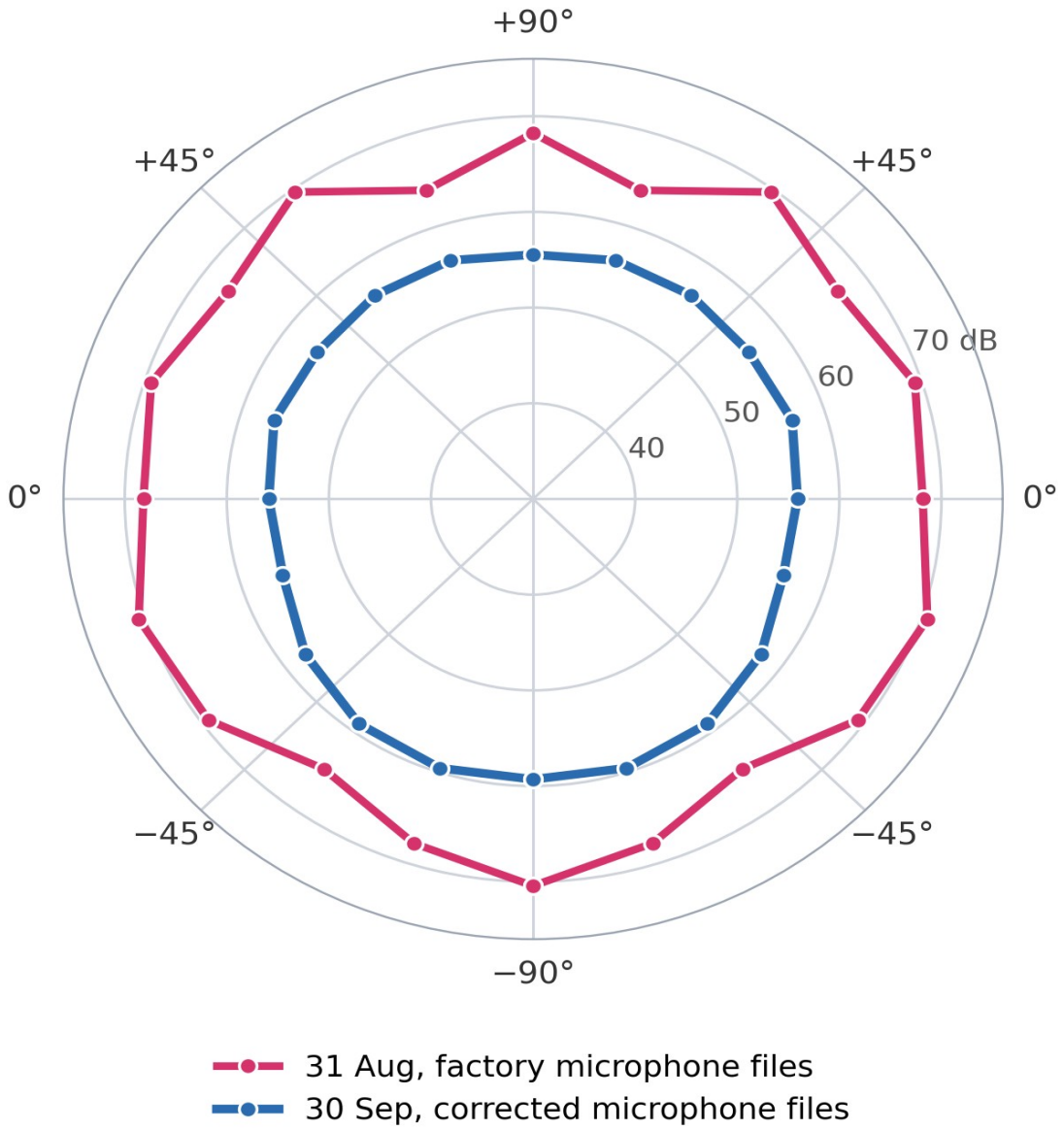
0 at the limit 2× or more

Worst microphone as a share of the limit. Cells: green 100 %, yellow 50 %, red 0 %.

Bands clearly over (of 13): **7 → 6**

Cells in limit (13 bands): **60 % → 66 %**

* Partial bands: only the tones above 5 kHz are measured (4 and 5 tones). 3–5 kHz is left out because of the loudspeaker: its cone has a rocking mode near 4.4 kHz. ISO values are an analogue, not a qualification of the room.



COMPARISON

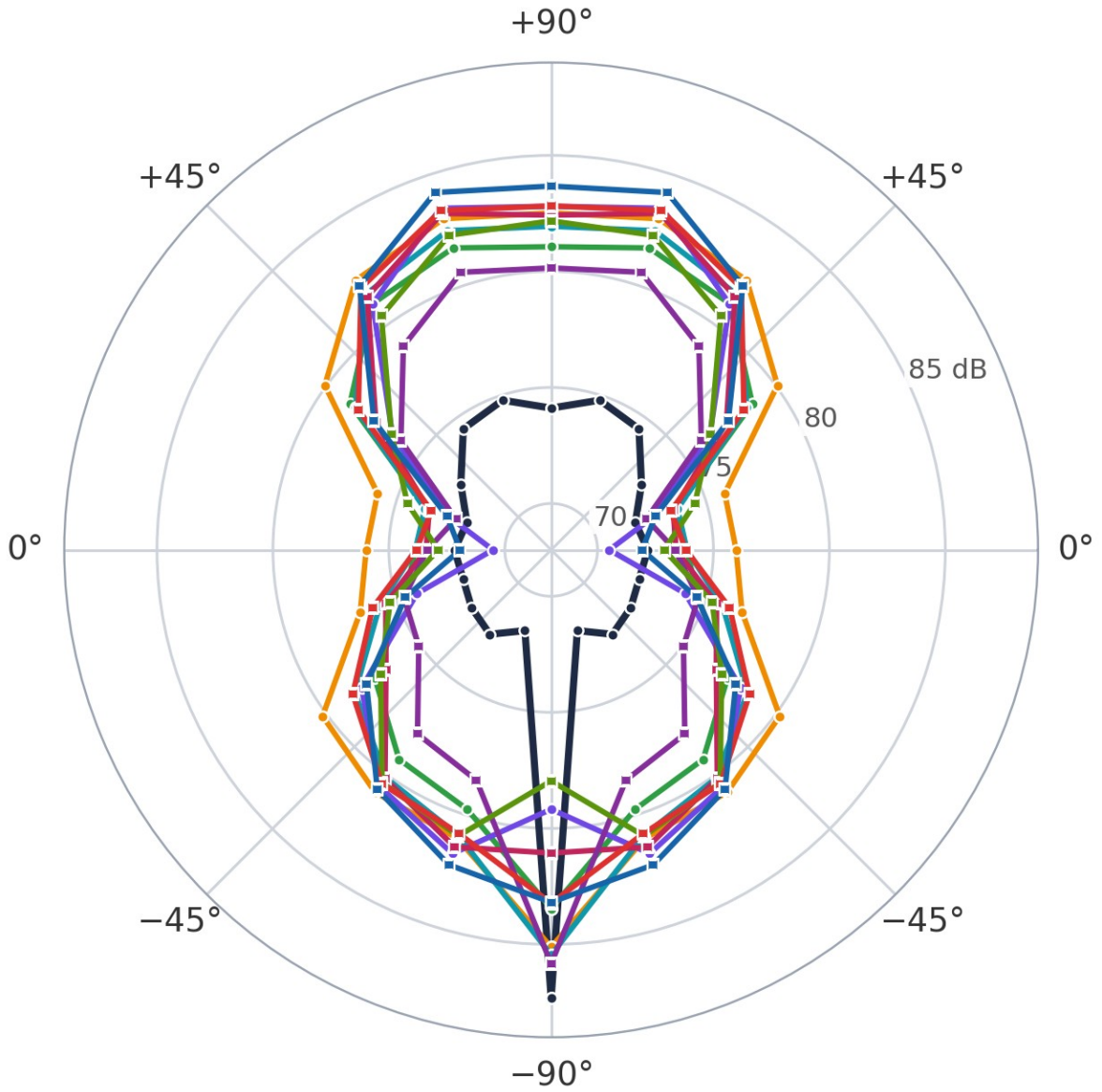
31 Aug and 30 Sep on the same axes

- Red: 31 Aug, factory microphone files. Blue: 30 Sep, corrected files. Two different propellers.
- The propeller was mounted the other way round between the two, so the thrust points the other way. We do not expect this to change the sound.
- **The 30 Sep outline is smoother.**

6.7 → 4.0 dB

between the loudest and quietest microphone

20–560 Hz, motor at 2000 µs. Different propellers and run conditions, so the levels are not comparable; compare the shapes.



THE MATERIALS

Materials at 2000 µs

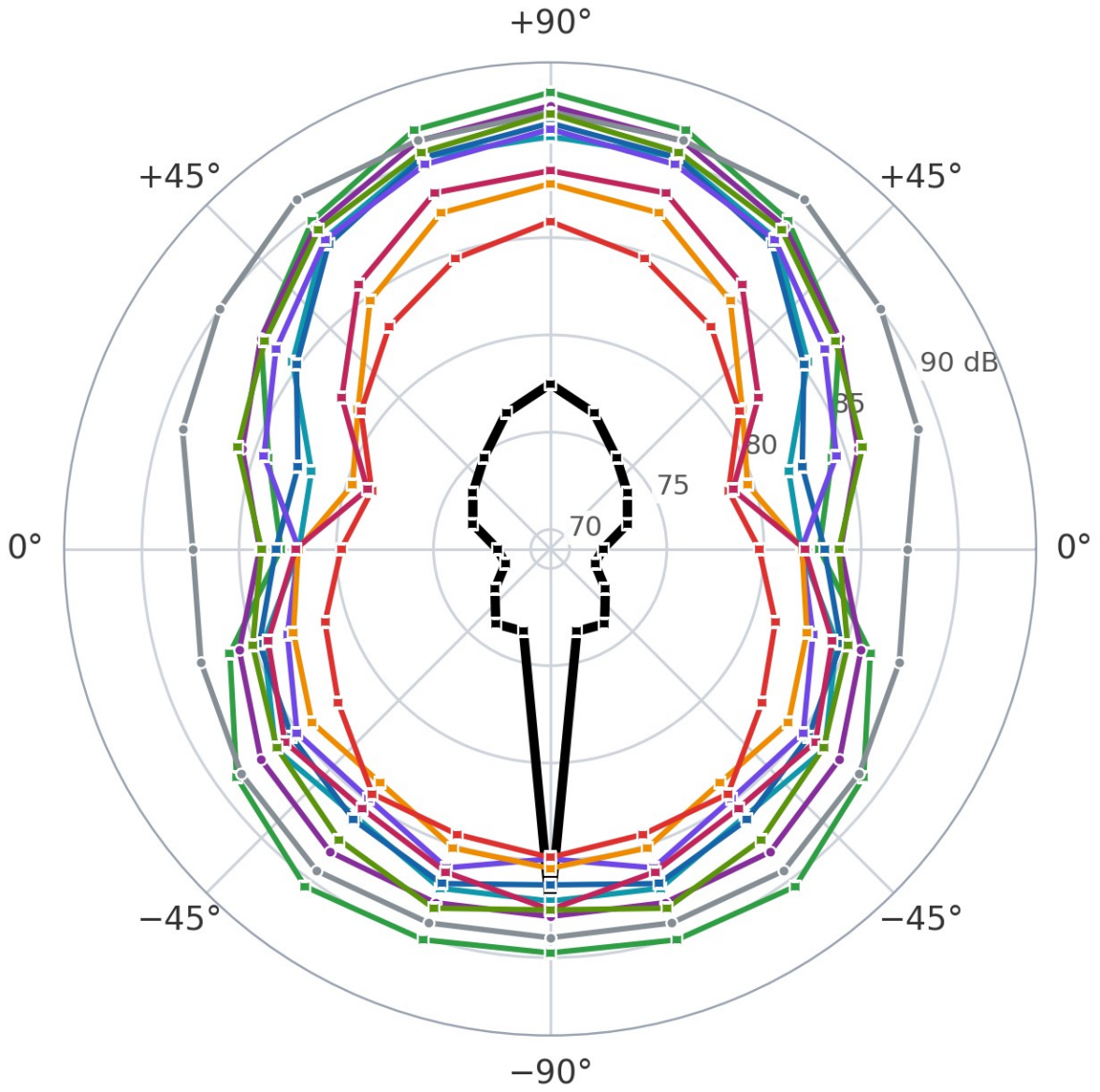
Mean level. Round markers: 2 Oct. Square: 7 Oct.

Bare	2 Oct	74.2 dB
PU foam	7 Oct	77.6 dB
Felt + rubber	2 Oct	78.2 dB
PU foam, double layer	7 Oct	78.2 dB
Cork	2 Oct	78.6 dB
Inner felt + outer PU foam	7 Oct	78.8 dB
Thinsulate, 8 layers	7 Oct	79.2 dB
Felt	2 Oct	79.3 dB
Thinsulate, 4 layers	7 Oct	79.4 dB
Rubber	2 Oct	80.3 dB

+3.4 to +6.2 dB

louder than bare, on average (bare measured 2 Oct)

2 and 7 Oct 2026, 5-inch 3-blade propeller, total level 500 Hz–24 kHz, motor at 2000 µs, corrected microphone files. Radial axis 68–89 dB.



THE MEMBRANES

Membranes at 2000 µs

Mean level. Round markers: 8 Oct. Square: 9 Oct.

Bare prop	9 Oct	75.1 dB
Thinsulate, 4 layers	9 Oct	82.8 dB
Thinsulate, 2 layers	9 Oct	84.0 dB
Thinsulate, 8 layers	9 Oct	85.0 dB
Felt + rubber	9 Oct	86.0 dB
Felt	9 Oct	86.3 dB
Felt, double layer	9 Oct	86.4 dB
Duct + PU foam, double	9 Oct	87.5 dB
PU foam	8 Oct	87.9 dB
Cork	9 Oct	88.5 dB
Duct, membrane only	8 Oct	89.5 dB

up to 10.2 dB

best against worst membrane set-up (Thinsulate, 4 layers and Duct, membrane only), at +18°; 6.7 dB on average

Thrust at 2000 µs: bare prop 3.75 N. The quietest other set-up, Thinsulate 4 layers, only 0.39 N.

8 and 9 Oct 2026, 5-inch 3-blade propeller, 500 Hz–24 kHz, 2000 µs, mean of 3 repeats, corrected files. Axis 69–94 dB. Not comparable with slide 8.